

# CHAPTER 2

## Estimation

### 2.1 Linear Model

Suppose we want to model the response  $Y$  in terms of three predictors,  $X_1$ ,  $X_2$  and  $X_3$ . One very general form for the model would be:

$$Y = f(X_1, X_2, X_3) + \epsilon$$

where  $f$  is some unknown function and  $\epsilon$  is the error in this representation.  $\epsilon$  is additive in this instance, but could enter in some more general form. Still, if we assume that  $f$  is a smooth, continuous function, that still leaves a very wide range of possibilities. Even with just three predictors, we typically will not have enough data to try to estimate  $f$  directly. So we usually have to assume that it has some more restricted form, perhaps linear as in:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

where  $\beta_i$ ,  $i=0, 1, 2, 3$  are unknown *parameters*.  $\beta_0$  is called the *intercept* term. Thus the problem is reduced to the estimation of four parameters rather than the infinite dimensional  $f$ . In a linear model the *parameters enter linearly*—the predictors themselves do not have to be linear. For example:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 \log X_2 + \beta_3 X_1 X_2 + \epsilon$$

is a linear model, but:

$$Y = \beta_0 + \beta_1 X_1^{\beta_2} + \epsilon$$

is not. Some relationships can be transformed to linearity—for example,  $y = \beta_0 x_1^{\beta_1} \epsilon$  can be linearized by taking logs. Linear models seem rather restrictive, but because the predictors can be transformed and combined in any way, they are actually very flexible. The term linear is often used in everyday speech as almost a synonym for simplicity. This gives the casual observer the impression that linear models can only handle small simple datasets. This is far from the truth—linear models can easily be expanded and modified to handle complex datasets. Linear is also used to refer to straight lines, but linear models can be curved. Truly nonlinear models are rarely absolutely necessary and most often arise from a theory about the relationships between the variables, rather than an empirical investigation.

## 2.2 Matrix Representation

If we have a response  $Y$  and three predictors,  $X_1$ ,  $X_2$  and  $X_3$ , the data might be presented in tabular form like this:

$$\begin{array}{cccc} y_1 & x_{11} & x_{12} & x_{13} \\ y_2 & x_{21} & x_{22} & x_{23} \\ \dots & & \dots & \\ y_n & x_{n1} & x_{n2} & x_{n3} \end{array}$$

where  $n$  is the number of observations, or *cases*, in the dataset.

Given the actual data values, we may write the model as:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i, i=1, \dots, n$$

but the use of subscripts becomes inconvenient and conceptually obscure. We will find it simpler both notationally and theoretically to use a matrix/vector representation. The regression equation is written as:

$$y = X\beta + \varepsilon$$

where  $y = (y_1, \dots, y_n)^T$ ,  $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)^T$ ,  $\beta = (\beta_0, \dots, \beta_3)^T$  and:

$$X = \begin{pmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ \dots & & \dots & \\ 1 & x_{n1} & x_{n2} & x_{n3} \end{pmatrix}$$

The column of ones incorporates the intercept term. One simple example is the *null model* where there is no predictor and just a mean  $y = \mu + \varepsilon$ :

$$\begin{pmatrix} y_1 \\ \dots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 \\ \dots \\ 1 \end{pmatrix} \mu + \begin{pmatrix} \varepsilon_1 \\ \dots \\ \varepsilon_n \end{pmatrix}$$

We can assume that  $E\varepsilon = 0$  since if this were not so, we could simply absorb the nonzero expectation for the error into the mean  $\mu$  to get a zero expectation.

## 2.3 Estimating $\beta$

The regression model,  $y = X\beta + \varepsilon$ , partitions the response into a systematic component  $X\beta$  and a random component  $\varepsilon$ . We would like to choose  $\beta$  so that the systematic part explains

as much of the response as possible. Geometrically speaking, the response lies in an  $n$ -dimensional space, that is,  $y \in \mathbb{R}^n$  while  $\beta \in \mathbb{R}^p$  where  $p$  is the number of parameters. If we include the intercept then  $p$  is the number of predictors plus one. We will use this definition of  $p$  from now on. It is easy to get confused as to whether  $p$  is the number of predictors or parameters, as different authors use different conventions, so be careful.

The problem is to find  $\beta$  so that  $X\beta$  is as close to  $Y$  as possible. The best choice, the estimate  $\hat{\beta}$ , is apparent in the geometrical representation seen in Figure 2.1.

$\hat{\beta}$  is, in this sense, the best estimate of  $\beta$  within the model space. The response predicted by the model is  $\hat{y} = X\hat{\beta}$  or  $Hy$  where  $H$  is an orthogonal projection matrix. The difference

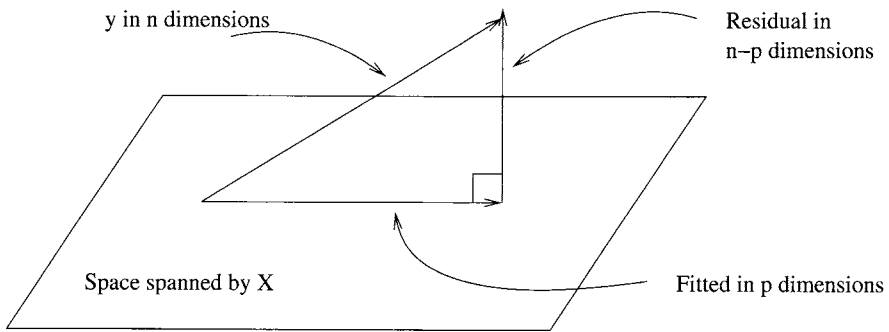


Figure 2.1 *Geometrical representation of the estimation  $\beta$ . The data vector  $Y$  is projected orthogonally onto the model space spanned by  $X$ . The fit is represented by projection  $\hat{y} = X\hat{\beta}$  with the difference between the fit and the data represented by the residual vector  $\hat{\epsilon}$*

between the actual response and the predicted response is denoted by  $\hat{\epsilon}$  and is called the *residuals*.

The conceptual purpose of the model is to represent, as accurately as possible, something complex,  $y$ , which is  $n$ -dimensional, in terms of something much simpler, the model, which is  $p$ -dimensional. Thus if our model is successful, the structure in the data should be captured in those  $p$  dimensions, leaving just random variation in the residuals which lie in an  $(n-p)$ -dimensional space. We have:

$$\begin{aligned} \text{Data} &= \text{Systematic Structure} + \text{Random Variation} \\ n \text{ dimensions} &= p \text{ dimensions} + (n-p) \text{ dimensions} \end{aligned}$$

## 2.4 Least Squares Estimation

The estimation of  $\beta$  can also be considered from a nongeometrical point of view. We might define the best estimate of  $\beta$  as the one which minimizes the sum of the squared errors. The *least squares* estimate of  $\beta$ , called  $\hat{\beta}$  minimizes:

$$\sum \epsilon_i^2 = \epsilon^T \epsilon = (y - X\beta)^T (y - X\beta)$$

Differentiating with respect to  $\beta$  and setting to zero, we find that  $\hat{\beta}$  satisfies:

$$X^T X \hat{\beta} = X^T y$$

These are called the *normal equations*. We can derive the same result using the geometrical approach. Now provided  $X^T X$  is invertible:

$$\begin{aligned}\hat{\beta} &= (X^T X)^{-1} X^T y \\ X \hat{\beta} &= X (X^T X)^{-1} X^T y \\ \hat{y} &= Hy\end{aligned}$$

$H = X(X^T X)^{-1} X^T$  is called the *hat-matrix* and is the orthogonal projection of  $y$  onto the space spanned by  $X$ .  $H$  is useful for theoretical manipulations, but you usually do not want to compute it explicitly, as it is an  $n \times n$  matrix which could be uncomfortably large for some datasets. The following useful quantities can now be represented using  $H$ .

The predicted or fitted values are  $\hat{y} = Hy = X\hat{\beta}$  while the residuals are  $\hat{\epsilon} = y - X\hat{\beta} = y - \hat{y} = (I - H)y$ . The residual sum of squares (RSS) is  $\hat{\epsilon}^T \hat{\epsilon} = y^T (I - H)^T (I - H)y = y^T (I - H)y$ .

Later, we will show that the least squares estimate is the best possible estimate of  $\beta$  when the errors  $\epsilon$  are uncorrelated and have equal variance or more briefly put  $\text{var } \epsilon = \sigma^2 I$ .

$\hat{\beta}$  is unbiased and has variance  $(X^T X)^{-1} \sigma^2$  provided  $\text{var } \epsilon = \sigma^2 I$ . Since  $\hat{\beta}$  is a vector, its variance is a matrix.

We also need to estimate  $\sigma^2$ . We find that  $E \hat{\epsilon}^T \hat{\epsilon} = \sigma^2 (n - p)$  which suggests the estimator:

$$\hat{\sigma}^2 = \frac{\hat{\epsilon}^T \hat{\epsilon}}{n - p} = \frac{\text{RSS}}{n - p}$$

as an unbiased estimate of  $\sigma^2$ .  $n - p$  is the *degrees of freedom* of the model. Sometimes you need the standard error for a particular component of  $\hat{\beta}$  which can be picked out as  $(\hat{\beta}_{i-1}) = \sqrt{(X^T X)^{-1}_{ii}} \hat{\sigma}$

## 2.5 Examples of Calculating $\hat{\beta}$

In a few simple models, it is possible to derive explicit formulae for  $\hat{\beta}$

1. When  $y = \mu + \varepsilon$ ,  $X = \mathbf{1}$  and  $\beta = \mu$  so  $X^T X = \mathbf{1}^T \mathbf{1} = n$  so:

$$\hat{\beta} = (X^T X)^{-1} X^T y = \frac{1}{n} \mathbf{1}^T y = \bar{y}$$

2. Simple linear regression (one predictor):

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\begin{pmatrix} y_1 \\ \dots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_1 \\ \dots & \dots \\ 1 & x_n \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \dots \\ \varepsilon_n \end{pmatrix}$$

We can now apply the formula but a simpler approach is to rewrite the equation as:

$$y_i = \overbrace{\beta_0}^{\beta'_0} + \beta_1 \bar{x} + \beta_1 (x_i - \bar{x}) + \varepsilon_i$$

so now:

$$X = \begin{pmatrix} 1 & x_1 - \bar{x} \\ \dots & \dots \\ 1 & x_n - \bar{x} \end{pmatrix} \quad X^T X = \begin{pmatrix} n & 0 \\ 0 & \sum_{i=1}^n (x_i - \bar{x})^2 \end{pmatrix}$$

Next work through the rest of the calculation to reconstruct the familiar estimates, that is:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x}) y_i}{\sum (x_i - \bar{x})^2}$$

In higher dimensions, it is usually not possible to find such explicit formulae for the parameter estimates unless  $X^T X$  happens to be a simple form. So typically we need computers to fit such models. Regression has a long history, so in the time before computers became readily available, fitting even quite simple models was a tedious time consuming task. When computing was expensive, data analysis was limited. It was designed to keep calculations to a minimum and restrict the number of plots. This mindset remained in statistical practice for some time even after computing became widely and cheaply available. Now it is a simple matter to fit a multitude of models and make more plots than one could reasonably study. The challenge for the analyst is to choose among these intelligently to extract the crucial information in the data.

## 2.6 Gauss-Markov Theorem

$\hat{\beta}$  is a plausible estimator, but there are alternatives. Nonetheless, there are three good reasons to use least squares:

1. It results from an orthogonal projection onto the model space. It makes sense geometrically.
2. If the errors are independent and identically normally distributed, it is the maximum likelihood estimator. Loosely put, the maximum likelihood estimate is the value of  $\beta$  that maximizes the probability of the data that was observed.
3. The Gauss-Markov theorem states that  $\hat{\beta}$  is the best linear unbiased estimate (BLUE).

To understand the Gauss-Markov theorem we first need to understand the concept of an *estimable function*. A linear combination of the parameters  $\psi = c^T \beta$  is estimable if and only if there exists a linear combination  $a^T y$  such that:

$$E a^T y = c^T \beta \quad \forall \beta$$

Estimable functions include predictions of future observations, which explains why they are well worth considering. If  $X$  is of full rank, then all linear combinations are estimable.

Suppose  $E\varepsilon=0$  and  $\text{var } \varepsilon=\sigma^2 I$ . Suppose also that the structural part of the model,  $EY=X\beta$  is correct. (Clearly these are big assumptions and so we will address the implications of this later.)  $\psi = c^T \beta$  be an estimable function; then the Gauss-Markov theorem states that in the class of all unbiased linear estimates of  $\psi$ ,  $\hat{\psi} = c^T \hat{\beta}$  has the minimum variance and is unique.

We prove this theorem. Suppose  $a^T y$  is some unbiased estimate of  $c^T \beta$  so that:

$$\begin{aligned} E a^T y &= c^T \beta & \forall \beta \\ a^T X \beta &= c^T \beta & \forall \beta \end{aligned}$$

which means that  $a^T X = c^T$ . This implies that  $c$  must be in the range space of  $X^T$  which in turn implies that  $c$  is also in the range space of  $X^T X$  which means there exists a  $\lambda$ , such that  $c = X^T X \lambda$  so:

$$c^T \hat{\beta} = \lambda^T X^T X \hat{\beta} = \lambda^T X^T y$$

Now we can show that the least squares estimator has the minimum variance—pick an arbitrary estimate  $a^T y$  and compute its variance:

$$\begin{aligned}
\text{var}(a^T y) &= \text{var}(a^T y - c^T \hat{\beta} + c^T \hat{\beta}) \\
&= \text{var}(a^T y - \lambda^T X^T y + c^T \hat{\beta}) \\
&= \text{var}(a^T y - \lambda^T X^T y) + \text{var}(c^T \hat{\beta}) + 2\text{cov}(a^T y - \lambda^T X^T y, \lambda^T X^T y)
\end{aligned}$$

but

$$\begin{aligned}
\text{cov}(a^T y - \lambda^T X^T y, \lambda^T X^T y) &= (a^T - \lambda^T X^T) \sigma^2 I X \lambda \\
&= (a^T X - \lambda^T X^T X) \sigma^2 I \lambda \\
&= (c^T - c^T) \sigma^2 I \lambda = 0
\end{aligned}$$

so

$$\text{var}(a^T y) = \text{var}(a^T y - \lambda^T X^T y) + \text{var}(c^T \hat{\beta})$$

Now since variances cannot be negative, we see that:

$$\text{var}(a^T y) \geq \text{var}(c^T \hat{\beta})$$

In other words,  $c^T \hat{\beta}$  has minimum variance. It now remains to show that it is unique. There will be equality in the above relationship if  $\text{var}(a^T y - \lambda^T X^T y) = 0$  which would require that  $a^T - \lambda^T X^T = 0$  which means that  $a^T y = \lambda^T X^T y = c^T \hat{\beta}$ . So equality occurs only if  $a^T y = c^T \hat{\beta}$  so the estimator is unique. This completes the proof.

The Gauss-Markov theorem shows that the least squares estimate  $\hat{\beta}$  is a good choice, but it does require that the errors are uncorrelated and have unequal variance. Even if the errors behave, but are nonnormal, then nonlinear or biased estimates may work better. So this theorem does not tell one to use least squares all the time; it just strongly suggests it unless there is some strong reason to do otherwise. Situations where estimators other than ordinary least squares should be considered are:

1. When the errors are correlated or have unequal variance, generalized least squares should be used. See Section 6.1.
2. When the error distribution is long-tailed, then robust estimates might be used. Robust estimates are typically not linear in  $y$ . See Section 6.4.
3. When the predictors are highly correlated (collinear), then biased estimators such as ridge regression might be preferable. See Chapter 9.

## 2.7 Goodness of Fit

It is useful to have some measure of how well the model fits the data. One common choice is  $R^2$ , the so-called *coefficient of determination* or *percentage of variance explained*:

$$R^2 = 1 - \frac{\sum(\hat{y}_i - y_i)^2}{\sum(y_i - \bar{y})^2} = 1 - \frac{\text{RSS}}{\text{Total SS(Corrected for Mean)}}$$

Its range is  $0 \leq R^2 \leq 1$ —values closer to 1 indicating better fits. For simple linear regression  $R^2 = r^2$  where  $r$  is the correlation between  $x$  and  $y$ . An equivalent definition

$$R^2 = \frac{\sum(\hat{y}_i - \bar{y})^2}{\sum(y_i - \bar{y})^2}$$

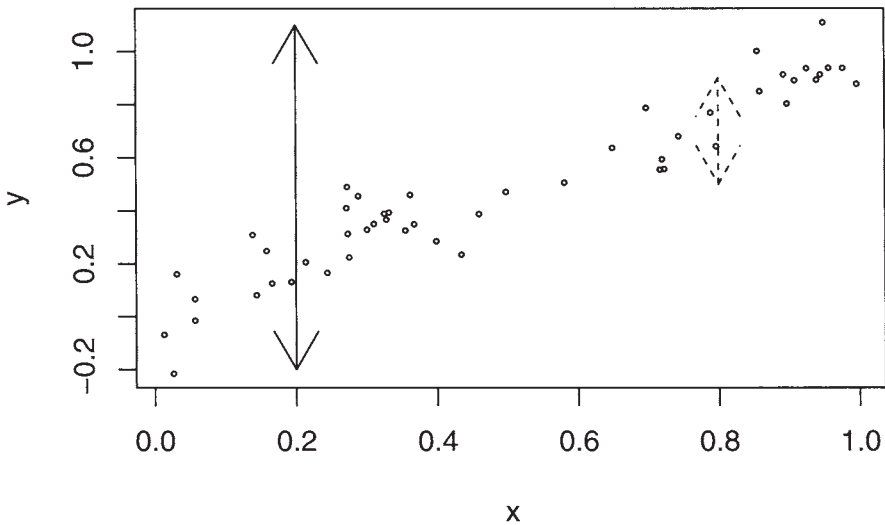


Figure 2.2 *Variation in the response  $y$  when  $x$  is known is denoted by dotted arrows while variation in  $y$  when  $x$  is unknown is shown with the solid arrows.*

The graphical intuition behind  $R^2$  is seen in Figure 2.2. Suppose you want to predict  $y$ . If you do not know  $x$ , then your best prediction is  $\bar{y}$  but the variability in this prediction is high. If you do know  $x$ , then your prediction will be given by the regression fit. This prediction will be less variable provided there is some relationship between  $x$  and  $y$ .  $R^2$  is one minus the ratio of the sum of squares for these two predictions. Thus for perfect predictions the ratio will be zero and  $R^2$  will be one.

$R^2$  as defined here does not make any sense if you do not have an intercept in your model. This is because the denominator in the definition of  $R^2$  has a null model with an intercept in mind when the sum of squares is calculated. Alternative definitions of  $R^2$  are possible when there is no intercept, but the same graphical intuition is not available and

the  $R^2$ s obtained in this way should not be compared to those for models with an intercept. Beware of high  $R^2$ s reported from models without an intercept.

What is a good value of  $R^2$ ? It depends on the area of application. In the biological and social sciences, variables tend to be more weakly correlated and there is a lot of noise. We would expect lower values for  $R^2$  in these areas—a value of 0.6 might be considered good. In physics and engineering, where most data come from closely controlled experiments, we expect to get much higher  $R^2$ s and a value of 0.6 would be considered low. Of course, I generalize excessively here so some experience with the particular area is necessary for you to judge your  $R^2$ s well.

An alternative measure of fit is  $\hat{\sigma}$ . This quantity is directly related to the standard errors of estimates of  $\beta$  and predictions. The advantage is that  $\hat{\sigma}$  is measured in the units of the response and so may be directly interpreted in the context of the particular dataset. This may also be a disadvantage in that one must understand whether the practical significance of this measure whereas  $R^2$ , being unitless, is easy to understand.

## 2.8 Example

Now let's look at an example concerning the number of species of tortoise on the various Galápagos Islands. There are 30 cases (Islands) and seven variables in the dataset. We start by reading the data into R and examining it (remember you first need to load the book data with the library (faraway) command):

```
> data (gala) >
gala
```

|           | Species | Endemics | Area  | Elevation | Nearest | Scruz |
|-----------|---------|----------|-------|-----------|---------|-------|
| Baltra    | 58      | 23       | 25.09 | 346       | 0.6     | 0.6   |
| Bartolome | 31      | 21       | 1.24  | 109       | 0.6     | 26.3  |
| . . .     |         |          |       |           |         |       |

The variables are Species—the number of species of tortoise found on the island, Endemics—the number of endemic species, Area—the area of the island ( $\text{km}^2$ ), Elevation—the highest elevation of the island (m), Nearest—the distance from the nearest island (km), Scruz—the distance from Santa Cruz Island (km), Adjacent—the area of the adjacent island ( $\text{km}^2$ ).

The data were presented by Johnson and Raven (1973) and also appear in Weisberg (1985). I have filled in some missing values for simplicity (see Chapter 12 for how this can be done). Fitting a linear model in R is done using the `lm ( )` command. Notice the syntax for specifying the predictors in the model. This is part of the *Wilkinson-Rogers* notation. In this case, since all the variables are in the gala data frame, we must use the `data=` argument

```
> mdl <- lm (Species ~ Area + Elevation + Nearest + Scruz
+ Adjacent, data=gala) >
summary (mdl)
```

Call:

```
lm (formula = Species ~ Area + Elevation + Nearest + Scruz
    + Adjacent, data = gala)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -111.68 | -34.90 | -7.86  | 33.46 | 182.58 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t ) |
|-------------|----------|------------|---------|----------|
| (Intercept) | 7.06822  | 19.15420   | 0.37    | 0.7154   |
| Area        | -0.02394 | 0.02242    | -1.07   | 0.2963   |
| Elevation   | 0.31946  | 0.05366    | 5.95    | 3.8e-06  |
| Nearest     | 0.00914  | 1.05414    | 0.01    | 0.9932   |
| Scruz       | -0.24052 | 0.21540    | -1.12   | 0.2752   |
| Adjacent    | -0.07480 | 0.01770    | -4.23   | 0.0003   |

Residual standard error: 61 on 24 degrees of freedom  
 Multiple R-Squared: 0.766, Adjusted R-squared: 0.717  
 F-statistic: 15.7 on 5 and 24 DF, p-value: 6.84e-07

We can identify several useful quantities in this output. Other statistical packages tend to produce output quite similar to this. One useful feature of R is that it is possible to directly calculate quantities of interest. Of course, it is not necessary here because the `lm()` function does the job, but it is very useful when the statistic you want is not part of the prepackaged functions.

First, we make the  $X$ -matrix:

```
> x <- model.matrix ( ~ Area + Elevation + Nearest + Scruz
    + Adjacent, gala)
```

and here is the response  $y$ :

```
> y <- gala$Species
```

Now let's construct  $(X^T X)^{-1}$ . `t()` does transpose and `%*%` does matrix multiplication. `solve(A)` computes  $A^{-1}$  while `solve(A, b)` solves  $Ax=b$ :

```
> xtxi <- solve (t (x) %*% x)
```

We can get  $\hat{B}$  directly, using  $(X^T X)^{-1} X^T y$ :

```
> xtxi %*% t (x) %*% y
      [,1]
1      7.068221
Area    -0.023938
Elevation 0.319465
Nearest  0.009144
Scruz    -0.240524
Adjacent -0.074805
```

This is a very bad way to compute  $\hat{\beta}$ . It is inefficient and can be very inaccurate when the predictors are strongly correlated. Such problems are exacerbated by large datasets. A better, but not perfect, way is:

```
> solve (crossprod (x, x), crossprod (x, y))
      [,1]
1      7.068221
Area    -0.023938
Elevation 0.319465
Nearest  0.009144
Scruz    -0.240524
Adjacent -0.074805
```

where `crossprod (x, y)` computes  $x^T y$ . Here we get the same result as `lm ( )` because the data are well-behaved. In the long run, you are advised to use carefully programmed code such as found in `lm ( )`. To see the full details, consult a text such as Thisted (1988).

We can extract the regression quantities we need from the model object. Commonly used are `residuals ( )`, `fitted ( )`, `deviance ( )` which gives the RSS, `df . residual ( )` which gives the degrees of freedom and `coef ( )` which gives the  $\hat{\beta}$ . You can also extract other needed quantities by examining the model object and its summary:

```
> names (mdl)
[1] "coefficients" "residuals"      "effects"
[4] "rank"         "fitted.values"  "assign"
[7] "qr"          "df.residual"    "xlevels"
[10] "call"        "terms"         "model"
> mdls <- summary (mdl)
> names (mdls)
[1] "call"        "terms"         "residuals"
[4] "coefficients" "aliases"       "sigma"
[7] "df"          "r.squared"     "adj.r.squared"
[10] "fstatistic"  "cov.unscaled"
```

We can estimate  $\sigma$  using the formula in the text above or extract it from the summary object:

```
> sqrt (deviance (mdl) /df.residual (mdl))
[1] 60.975
> mdls$sigma
[1] 60.975
```

We can also extract  $(X^T X)^{-1}$  and use it to compute the standard errors for the coefficients. (`diag ( )` returns the diagonal of a matrix):

```
> xtxi <- mdls$cov.unscaled
> sqrt (diag (xtxi)) *60.975
(Intercept)      Area      Elevation      Nearest      Scruz
 19.154139  0.022422    0.053663    1.054133    0.215402
```

```
Adjacent
0.017700
```

or get them from the summary object:

```
> mdl$coef [,2]
(Intercept)      Area  Elevation  Nearest      Scrz
 19.154198   0.022422   0.053663   1.054136   0.215402
Adjacent
0.017700
```

Finally, we may compute or extract  $R^2$ :

```
> 1-deviance (mdl) /sum ((y-mean (y))^2)
[1] 0.76585
> mdl$sr.squared
[1] 0.76585
```

## 2.9 Identifiability

The least squares estimate is the solution to the normal equations:

$$X^T X \hat{\beta} = X^T y$$

where  $X$  is an  $n \times p$  matrix. If  $X^T X$  is singular and cannot be inverted, then there will be infinitely many solutions to the normal equations and  $\hat{\beta}$  is at least partially unidentifiable. Unidentifiability will occur when  $X$  is not of full rank—when its columns are linearly dependent. With observational data, unidentifiability is usually caused by some oversight. Here are some examples:

1. A person's weight is measured both in pounds and kilos and both variables are entered into the model.
2. For each individual we record the number of years of preuniversity education, the number of years of university education and also the total number of years of education and put all three variables into the model.
3. We have more variables than cases, that is,  $p > n$ . When  $p = n$ , we may perhaps estimate all the parameters, but with no degrees of freedom left to estimate any standard errors or do any testing. Such a model is called *saturated*. When  $p > n$ , then the model is called *supersaturated*. Oddly enough, such models are considered in large-scale screening experiments used in product design and manufacture, but there is no hope of uniquely estimating all the parameters in such a model.

Such problems can be avoided by paying attention. Identifiability is more of an issue in designed experiments. Consider a simple two-sample experiment, where the treatment observations are  $y_1, \dots, y_n$  and the controls are  $y_{n+1}, \dots, y_{m+n}$ . Suppose we try to model the response by an overall mean  $\mu$  and group effects  $\alpha_1$  and  $\alpha_2$ :

$$y_j = \mu + \alpha_i + \epsilon_j \quad i = 1, 2 \quad j = 1, \dots, m+n$$

$$\begin{pmatrix} y_1 \\ \vdots \\ y_n \\ y_{n+1} \\ \vdots \\ y_{m+n} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ & \dots & \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ \vdots & \vdots & \vdots \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \vdots \\ \vdots \\ \epsilon_{m+n} \end{pmatrix}$$

Now although  $X$  has three columns, it has only rank 2— $(\mu, \alpha_1, \alpha_2)$  are not identifiable and the normal equations have infinitely many solutions. We can solve this problem by imposing some constraints,  $\mu=0$  or  $\alpha_1+\alpha_2=0$ , for example.

Statistics packages handle nonidentifiability differently. In the regression case above, some may return error messages and some may fit models because rounding error may remove the exact identifiability. In other cases, constraints may be applied but these may be different from what you expect. By default, R fits the largest identifiable model by removing variables in the reverse order of appearance in the model formula.

Here is an example. Suppose we create a new variable for the Galápagos dataset—the difference in area between the island and its nearest neighbor:

```
> gala$Adiff <- gala$Area -gala$Adjacent
```

and add that to the model:

```
> g <- lm (Species ~ Area+Elevation+Nearest+Scruz+Adjacent
+Adiff, gala)
> summary (g)
Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  7.06822    19.15420   0.37    0.7154
Area         -0.02394     0.02242  -1.07    0.2963
Elevation     0.31946     0.05366   5.95 3.8e-06
Nearest       0.00914     1.05414   0.01  0.9932
Scruz         -0.24052     0.21540  -1.12  0.2752
Adjacent     -0.07480     0.01770  -4.23  0.0003
Adiff         NA          NA      NA      NA
Residual standard error: 61 on 24 degrees of freedom
Multiple R-Squared: 0.766, Adjusted R-squared: 0.717
F-statistic: 15.7 on 5 and 24 DF, p-value: 6.84e-07
```

We get a message about one undefined coefficient because the rank of the design matrix  $X$  is six, which is less than its seven columns. In most cases, the cause of identifiability can be revealed with some thought about the variables, but, failing that, an eigendecomposition of  $X^T X$  will reveal the linear combination(s) that gave rise to the unidentifiability.

Lack of identifiability is obviously a problem, but it is usually easy to identify and work around. More problematic are cases where we are close to unidentifiability. To demonstrate this, suppose we add a small random perturbation to the third decimal place of *Adiff* by adding a random variate from  $U[-0.005, 0.005]$  where  $U$  denotes the uniform distribution:

```
> Adiffe <- gala$Adiff+0.001*(runif(30)-0.5)
```

and now refit the model:

```
> g <- lm (Species ~ Area+Elevation+Nearest+Scruz
+Adjacent+Adiffe, gala)
> summary (g)
Coefficients:
              Estimate      Std. Error t value Pr(>|t|)
(Intercept)  7.14e+00      1.956e+01   0.37    0.72
Area        -2.38e+04      4.70e+04  -0.51    0.62
Elevation    3.12e-01      5.67e-02   5.50  1.46e-05
Nearest      1.38e-01      1.10e+00   0.13    0.90
Scruz        -2.50e-01      2.20e-01  -1.14    0.27
Adjacent     2.386e+04      4.70e+04   0.51    0.62
Adiffe       2.386e+04      4.70e+04   0.51    0.62
Residual standard error: 61.9 on 23 degrees of freedom
Multiple R-Squared: 0.768, Adjusted R-squared: 0.708
F-statistic: 12.7 on 6 and 23 DF, p-value: 2.58e-06
```

Notice that now all parameters are estimated, but the standard errors are very large because we cannot estimate them in a stable way. We deliberately caused this problem so we know the cause but in general we need to be able to identify such situations. We do this in Section 5.3.

## Exercises

- The dataset *teengamb* concerns a study of teenage gambling in Britain. Fit a regression model with the expenditure on gambling as the response and the sex, status, income and verbal score as predictors. Present the output.
  - What percentage of variation in the response is explained by these predictors?
  - Which observation has the largest (positive) residual? Give the case number.
  - Compute the mean and median of the residuals.
  - Compute the correlation of the residuals with the fitted values.
  - Compute the correlation of the residuals with the income.
  - For all other predictors held constant, what would be the difference in predicted expenditure on gambling for a male compared to a female?
- The dataset *uswages* is drawn as a sample from the Current Population Survey in 1988. Fit a model with weekly wages as the response and years of education and experience as predictors. Report and give a simple interpretation to the regression

coefficient for years of education. Now fit the same model but with logged weekly wages. Give an interpretation to the regression coefficient for years of education. Which interpretation is more natural?

3. In this question, we investigate the relative merits of methods for computing the coefficients. Generate some artificial data by:

```
> x <- 1:20
> y <- x+rnorm(20)
```

Fit a polynomial in  $x$  for predicting  $y$ . Compute  $\hat{\beta}$  in two ways—by `lm()` and by using the direct calculation described in the chapter. At what degree of polynomial does the direct calculation method fail? (Note the need for the `I()` function in fitting the polynomial, that is, `lm(y~x+I(x^2))`).

4. The dataset prostate comes from a study on 97 men with prostate cancer who were due to receive a radical prostatectomy. Fit a model with `lpsa` as the response and `l.cavol` as the predictor. Record the residual standard error and the  $R^2$ . Now add `lweight`, `svi`, `lpph`, `age`, `lcp`, `pgg45` and `gleason` to the model one at a time. For each model record the residual standard error and the  $R^2$ . Plot the trends in these two statistics.
5. Using the prostate data, plot `lpsa` against `l.cavol`. Fit the regressions of `lpsa` on `l.cavol` and `l.cavol` on `lpsa`. Display both regression lines on the plot. At what point do the two lines intersect?

